



Republic of Rwanda
Ministry of Education



LSVSL501

SURVEYING LABORATORY MAINTAINANCE

Maintain Surveying laboratory

Competence

RQF Level: 5

Learning Hours



Credits: 7

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV5002-TVET Certificate V in Land Surveying

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Purpose statement	This module describes the knowledge, skills and right attitude required to maintain surveying laboratory. At the end of this module the learner should be able to organize surveying lab, calibrate and repair surveying instruments.					
Leaning assumed to be in place	Applied physics, Applied mathematics					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Organize surveying lab	1.1. Surveying materials, tools, and equipment are regularly recorded according to their types.
	1.2. Safety measures are properly applied according to the established protocols and regulatory standards.
	1.3. Surveying materials, tools and equipment are well stored according to their types.
	1.4. Calibration and repairing tools and equipment are properly selected according to the type of work
2. Calibrate the instruments	2.1. Surveying instrument components are properly identified according to their types
	2.2. Surveying instrument status is properly checked according to the manufacturer's manual
	2.3. Surveying instrument is correctly adjusted according to the manufacturer's manual
3. Repair the instruments	3.1. Types of instrument defects are properly identified in accordance with the functionality of the instrument.
	3.2. The surveying instrument defects are correctly fixed in accordance with the functionality of the instrument.
	3.3. Repaired instrument is properly test in accordance with the functionality of the instrument.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describing principles of ethics and regulatory compliance ✓ Describing lab management principles ✓ Identifying lab maintenance principles ✓ Identifying surveying tools and equipment ✓ Describing components of surveying equipment structure ✓ Identifying calibration tools and equipment ✓ Identifying calibration methods ✓ Describing principles of safety, health and security at workplace	✓ Applying mathematics ✓ Applying calibration techniques ✓ Applying repairing skills ✓ Operating surveying equipment ✓ Applying managerial skills ✓ Applying maintenance skills ✓ Applying reporting skills ✓ Applying communication skills ✓ Applying ICT skills	✓ Having Critical Thinking attitude ✓ Being persistent ✓ Being honest ✓ Being Self-motivated ✓ Being decisive ✓ Being well- organized ✓ Being research oriented ✓ Being creative and innovative ✓ Being Flexible ✓ Being attentive ✓ Being Detail-oriented ✓ Being prompt ✓ Having teamwork spirit ✓ Being confident ✓ Being responsible ✓ Having problem solving attitude

Course content

Learning outcomes	At the end of the module the learner will be able to:
	1. Organize surveying lab 2. Calibrate the instruments 3. Repair the instruments

Learning outcome 1: Organize surveying lab

Learning hours: 15 hours

Indicative content

- **Recording Surveying materials, tools, and equipment**
 - ✓ Identifying materials, tools and equipment
 - ✓ Identifying types of documents used for recording
 -  Bin card
 -  Stores ledger
 -  Stores issue requisition
 -  Bill of material

- ✚ Material transfer note
- ✓ Applying Methods used in recording
 - ✚ Paper system
 - ✚ Electronic system
- ✓ Ensure record retention
- **Applying safety measures**
 - ✓ Safety measures for occupants
 - ✓ Safety measures for material, tools and equipment
 - ✓ Ensuring store room requirements
 - ✚ Ventilation
 - ✚ Fire protection
 - ✚ Lighting
 - ✚ Alarm
 - ✚ Camera system
- **Storing Surveying materials, tools and equipment**
 - ✓ Identify types of stores
 - ✚ Centralized stores
 - ✚ Decentralized stores
 - ✚ Imprest store
 - ✓ Apply storing procedures
 - ✚ Setting out safety preventive measures
 - ✚ Categorizing surveying materials, tools and equipment
 - ✚ Coding surveying materials, tools and equipment
 - ✚ Filling store documents
 - ✚ Positioning /placement of surveying materials, tools and equipment
- **Selecting calibration and repairing tools and equipment**
 - ✓ Tools
 - ✓ Equipment
 - ✓ Materials

Resources required for the indicative content

Equipment	spray gun, Compressor, PPE, drilling machine, Screw gun, Staple gun, ladder, total station, laser scanner, theodolites, EDMs, level, Computer, Projector, printer, measuring wheel, GNSS Receivers.
Materials	Paint, water, nails, screws, wooden glue, Beacons, warning tape, pegs, timber planks
Tools	Bucket, Tape measure, Paint brush, Laser meter, Scrapper, Spirit levels, Notebook, pencils Hack saw, Screw driver, Hand saw, Chisels, claw Hammer, screw drivers, pangas, ranging rods, staff readings, prism poles, tripods, reflectors.

Facilitation techniques	Group discussion, scenario, simulation, individual activity, practical works - Trainer displays a video on storing process and asks the learners to discuss on types of surveying materials, tools and equipment in manageable groups. Trainer asks learners to present their work and then trainer provides feedback and input and asks them to refine their work if necessary. - Trainer asks learners to categorize code, fill store document, and store surveying materials, tools and equipment in their correct place and position. Trainer checks if instruments and equipment are properly stored and then provide feedback.
Formative assessment methods	- Written assessment - Oral assessment - Performance assessment

Learning outcome 2: Calibrate the instruments	Learning hours: 25 hours
Indicative content	
Identifying surveying instrument components ✓ Theodolite ✓ Levels ✓ Total station ✓ EDM instruments ✓ GNSS Receivers ✓ Laser scanners Checking Surveying instrument status ✓ Checking functionality of surveying instruments ✚ Zero-point checks ✚ Linearity checks ✚ Calibration checks ✓ Elaborating status instrument report Adjusting surveying instrument ✓ Applying calibration methods ✚ Static calibration ✚ Dynamic calibration ✚ Comparative calibration ✚ Software calibration ✓ Conducting operational Checks ✓ Elaborating report	

Resources required for the indicative content	
Equipment	Total station, Levels, Theodolite, EDMs instruments, GNSS Receivers, Laser scanners, Projector, Collimator
Materials	Oil, Sand paper, water, nails, screws, pegs, Notebook
Tools	Brushes, pencils, screw driver, Claw Hammer, machete, ranging rods, staff readings, prism poles, tripods, reflectors, plumb bob and hub spot.
Facilitation techniques	Group discussion, simulation, demonstration, individual activity. Practical work • Trainer takes learners in the surveying lab equipped with surveying instruments out of calibration and shows them the video on how the statuses of surveying instruments are checked. Trainer asks them to check the status of the instruments and produce the instrument status report. • Trainer asks learners to prepare list of materials, tools, and equipment required to calibrate those instruments and trainer verifies the provided list. • Trainer demonstrates to the learner how calibration is done and asks them to calibrate the identified instrument out of calibration. Trainer monitors the activity and provides support where necessary.
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment

Learning outcome 3: Repair the instruments	Learning hours: 30 hours
Indicative content	
• Identifying surveying instrument defects ✓ Mechanical damage ✓ Optical defects ✓ Electronic malfunctions ✓ Software bugs ✓ Battery problem • Fixing instrument defect ✓ Applying safety rules and regulations ✓ Applying repairing procedures ✚ Firmware ✚ Hardware ✓ Elaborating repairing report	

- **Testing repaired instrument**

- ✓ Functionality tests
 - ✚ Optical Tests
 - ✚ Electronic and Software Tests
 - ✚ Measurement Tests
- ✓ Accessory Tests
 - ✚ Tripods
 - ✚ Prisms and Reflectors
 - ✚ Cables and Connectors
- ✓ Documentation and Reporting

Resources required for the indicative content

Equipment	spray gun, Compressor, PPE, drilling machine, Staple gun, total station, laser scanner, theodolites, EDMs, levels, Projector, Computer, printer, measuring wheel, GNSS Receivers.
Materials	Paint, water, timber, nails, screws, wooden glue, oil
Tools	Tape measure, Notebook, pencils, claw Hammer, screw drivers, tripods, staff readings, club hammer, prism pole, reflector, ranging rods, measuring tape
Facilitation techniques	Group discussion, scenario, simulation, individual activity • Trainer takes learners in lab of surveying instruments with and without defects and shows them a video on surveying instrument repairing process. Trainer asks the learners in their respective groups to analyse and discover the status of instruments then prepare the report on the way forward. • Trainer ask learner to apply methods and procedures for fixing defected surveying instruments. Trainer monitors the activity and provides support where necessary.
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment

Integrated / Summative assessment

Integrated situation

DT Survey Ltd located in GASABO District has received a donation from government of Rwanda for maintaining different surveying tools and equipment in its Laboratory. Now DT Survey Ltd would like to offer job for interested Lab technician in land surveying firm to maintain its new surveying lab stated. As you are technician of the EL3000 Ltd signed the contract; you are requested to carry out:

- i) Upgrading of laboratory setting,
- ii) Calibration of one total station,
- iii) Repairing of one tripod

Based on the emergency, the work is planned to be done within 10 hours.

Resources

Tools	6 staff readings, 12 tripods, 12 plumb bobs, brushes, Tool kit
Equipment	Total station, 6 levels, 6 theodolites, collimator
Materials/ Consumables	Paint, water, oil, timber, nails, screws, wooden glue, pegs

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Organize surveying lab 30%	1.1. Surveying materials, tools, and equipment are well recorded	Format requirements is respected			4
		Recording method applied			3
	1.2. Safety measures are well applied	PPEs are worn			3
		Tools and equipment are safely used			3
	1.3. Surveying materials, tools and equipment are well stored	Preventive measures are set			4
		Tools and equipment are categorized			3
		Required tools and equipment are coded			5
		Tools and equipment are placed and positioned			5
Learning outcome 2: Calibrate the instruments 40%	2.1. Surveying instrument components are properly identifying	Electronic theodolite component identified			3
		EDM component identified			3
		Microprocessor component is identified			3
		Data storage component is identified			2

		data display component is identified			2
	2.2. Surveying instrument status is well checked	Surveying instrument status report is elaborated			7
	2.3. Calibration tools and equipment are well selected	tools are selected			3
		Equipment are selected			3
		Materials are selected			3
	2.4. Surveying instrument is well adjusted	Surveying instrument is checked			5
		Surveying instrument adjustment Report is elaborated			6
Learning outcome 3: Repair the instruments 30%	3.1. The repairing materials are well selected	Instrument defects are identified			3
		Spare parts are selected			3
		Accessories are selected			3
	3.2. Repairing tools and equipment are properly selected	Repairing tools are selected			3
		Repairing equipment are selected			3
	3.3. Instrument defect is correctly fixed	Repairing procedures are applied			8
		Fixing instrument defect report is elaborated			7
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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